Seed, Envelope, Detail: Coarse-to-Fine Diffusion Attention for Training-Free Segmentation

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Abstract

Text-to-image diffusion models localize words in cross-attention (CA) and group pixels in self-attention (SA), so the images they generate can be segmented without training. We show that how these signals are combined across spatial scales *matters more than which graph operator is applied to them. Our training-free pipeline builds a class-word CA seed from the coarse-dominated modules, weighted by their agreement with SA. It then propagates the seed coarse to fine through SA at 16, 32 and 64 pixels. A one-sided cap lets each finer level redistribute evidence only within the mass its coarse parent assigns, and a fixed DenseCRF aligns the result with image edges. Each stage has a measurable role. Coarse SA supplies a high-recall object envelope that is robust to clutter, and fine SA restores detail. The cap removes 13% of background mass but only 0.5% of object mass, whereas exact mass conservation forces a wrong parent's mass onto its children and costs up to five points. DenseCRF gains six times more on capped maps in cluttered scenes. Selecting every choice on five class-stratified folds of VOC-Sim, the pipeline reaches 0.829 out-of-fold mIoU and transfers frozen to COCO-Cap at 0.696, against 0.704 and 0.582 for class-word OVAM. An oracle analysis shows that the hierarchy's gain is cross-image calibration of the propagated maps, which also explains its one systematic failure, objects that fill the frame.*

1. Introduction

A diffusion model that draws "a photograph of a sheep" must decide, at every denoising step, which image positions belong to the word *sheep* and which positions belong together. The first decision is visible in cross-attention (CA) between image queries and text keys; the second in self-attention (SA) between image positions. Reading both out yields a segmentation mask of the generated image without any training [6, 11], which makes generated images a source of free pixel labels.

Turning attention into masks is nevertheless a design problem: Stable Diffusion 1.5 exposes 31 denoising calls, four spatial scales, sixteen attention layers of each type and eight heads per layer. Existing pipelines average CA over modules, propagate it once with fine SA and refine the result [6]. Recent work adds re-weighting, graph sharpening or random walks on top of that recipe [4, 5, 7, 8]. We tested many such operators under a strict protocol, and most did not help. What did help was respecting the different roles that attention plays at different *scales* (Fig. 1). Coarse CA marks the object reliably and fine CA is noisy, so we weight CA layers by how well their spatial grouping agrees with self-attention. Coarse SA covers the whole object but blurs its boundary, and fine SA is precise but misses parts. We therefore propagate from coarse to fine and let each finer level redistribute evidence only *within* the mass its parent assigns, never beyond it (a one-sided cap). DenseCRF then aligns the result with image edges.

The resulting pipeline uses no training, no optimized tokens and no tuned graph parameters. On VOC-Sim it reaches 0.829 out-of-fold mIoU, and on COCO-Cap it reaches 0.696 in frozen transfer, against 0.704 and 0.582 for OVAM with class words [6]. Every choice is made on VOC-Sim folds, with COCO-Cap used only after freezing.

Our contributions are as follows.

- A training-free pipeline that turns diffusion attention into class masks. It weights cross-attention layers by their agreement with self-attention, propagates the resulting seed from coarse to fine self-attention, and caps each finer level by the coarser one.
- Controlled ablations that isolate each choice: CA scales and layer weighting, coarse versus fine SA, the one-sided cap versus exact mass conservation and no constraint, and DenseCRF on each kind of map. Each is backed by quantitative and qualitative evidence.
- An oracle threshold analysis showing that the hierarchy improves calibration across images rather than ranking within an image, which explains its main failure case.
- A fold-safe selection protocol, with every choice made on VOC-Sim folds and the negative results it produced

reported in full.

2. Related work

Attention maps as segmentations. DAAM aggregates diffusion CA over layers and steps to attribute image regions to prompt words [11]. OVAM adds a separate attribution prompt, so that any word can be queried, fuses CA with 64-pixel SA, refines with DenseCRF, and optionally optimizes tokens on annotated images [6]. It also released VOC-Sim and COCO-Cap, the benchmarks we use. We keep its class-word setting and its DenseCRF, and change how CA is aggregated and how SA is applied across scales.

Selecting and weighting attention. GoCA weights CA layers by the agreement between their features and a self-attention structure [7]; our self-attention-guided layer weighting adapts this idea to every denoising call and a class-word query. NERVE weights heads by semantic output and propagates with a stopped random walk [5]. We tested adaptations of both in our setting (supplementary material): the layer weighting improved the final mask, while the head weighting and the random walk did not.

Propagation over self-attention graphs. SA is a row-stochastic affinity, so repeated application spreads a seed and eventually leaks across boundaries. SPARK sharpens the affinity by expansion, inflation and pruning [4]. M2N2 treats SA as a Markov chain and uses arrival times [2]. PEARL studies attention geometry for training-free segmentation [9]. Multi-resolution use of SA appears in FA-Seg, which fuses refinements from several resolutions with fixed weights [1], and in SeeDiff, which re-extracts seeds between scales [8]. Our hierarchy instead passes one map from coarse to fine. Its key ingredient, the one-sided cap, makes the coarse level an upper bound on what the finer level may claim. We compare it directly with exact (two-sided) mass conservation.

Boundary refinement. DenseCRF combines a unary prior with appearance-dependent pairwise potentials [3]. Attention pipelines apply it with fixed parameters [6]; we do the same and analyze when it helps.

3. Method

Given an image generated by a text-to-image diffusion model and a class word w (*e.g.* "sheep"), we predict a binary mask of the pixels that belong to w. All signals are attention maps recorded while the image is generated, and no parameter is trained. The method has four steps (Fig. 1). First, cross-attention gives a class-word map in every attention layer (Sec. 3.2). Second, these maps are combined into a semantic seed, weighting each layer by how well its spatial structure agrees with self-attention (Sec. 3.3). Third, the seed is propagated through self-attention from coarse to fine resolution, with each finer level capped by the coarser one (Sec. 3.4). Fourth, the map is thresholded and refined with DenseCRF (Sec. 3.5).

3.1. Preliminaries: attention in latent diffusion

Stable Diffusion [10] generates an image in a latent space of size 64×64 (for a 512×512 image). Starting from Gaussian noise, a UNet denoises the latent over T steps; we use $T{=}30$ and record its 31 forward calls, indexed by t. The UNet contains transformer blocks at four spatial resolutions, $s \in \{64, 32, 16, 8\}$. At call t, block b receives image features $h_{b,t} \in \mathbb{R}^{n_b \times C_b}$, one row per latent position, where $n_b = s_b^2$. The prompt is encoded by the text encoder τ_θ into $X \in \mathbb{R}^{77 \times d_E}$, one row per token. Each block contains a self-attention layer followed by a cross-attention layer, and each layer has $H{=}8$ heads. For head h,

$$A_h = \mathrm{softmax}\Big(\frac{Q_h K_h^\top}{\sqrt{d}}\Big), \quad Q_h = h_{b,t} W_h^Q, \; K_h = Y W_h^K, \tag{1}$$

where W_h^Q, W_h^K are the layer's learned projections, d is the head dimension and the softmax runs over each row. The keys come from $Y = h_{b,t}$ in self-attention and from text in cross-attention. The two attention matrices therefore mean different things:

- **Cross-attention (CA)**, $A^{\mathrm{ca}}_{b,t,h} \in \mathbb{R}^{n_b \times 77}$. Row i is a distribution over text tokens for latent position i, so column k is a spatial map of where token k is attended [11].
- **Self-attention (SA)**, $A^{\mathrm{sa}}_{b,t,h} \in \mathbb{R}^{n_b \times n_b}$. Row i says from which positions position i gathers features. Rows sum to one, so for any map $x \in \mathbb{R}^{n_b}$ the product $A^{\mathrm{sa}} x$ replaces each value by an affinity-weighted average of the values at related positions. Repeated products therefore spread x over regions that the network groups together.

Following OVAM [6], the CA keys are computed from a separate *attribution prompt* X' ("a photograph of a [class]"), which contains w even when the generation prompt does not. We write $\bar{A}^s_t = \frac{1}{H}\sum_h A^{\mathrm{sa}}_{b,t,h}$ for the head-averaged SA matrix of the last decoder block at resolution s, and use U for bilinear $2\times$ upsampling and D for area downsampling.

3.2. Class-word cross-attention maps

For each CA layer b and call t, the class-word map averages the heads and the columns of all tokenizer pieces $k \in \mathcal{K}_w$ of w:

$$a_{b,t} = \frac{1}{H}\sum_{h=1}^{H} \frac{1}{|\mathcal{K}_w|} \sum_{k \in \mathcal{K}_w} A^{\mathrm{ca}}_{b,t,h}[:, k] \; \in \mathbb{R}^{n_b}. \tag{2}$$

We use the queries of the conditional (text-guided) branch. Stable Diffusion 1.5 has sixteen CA layers: five each at 64,

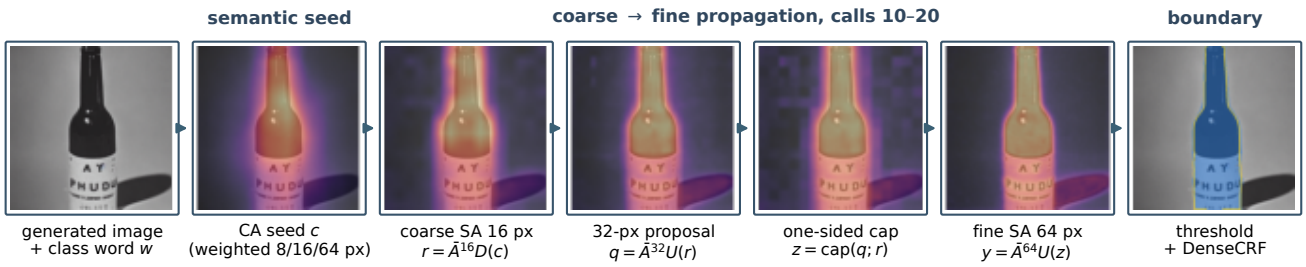


Figure 1. **Seed, envelope, detail, boundary.** Real intermediate maps for one VOC-Sim image. A class-word CA seed c (Sec. 3.3) is propagated by coarse 16-pixel SA, which spreads it over the whole bottle. The 32-pixel SA proposal q is then capped so that no 2×2 group exceeds its coarse parent's mass, which trims the background. Fine 64-pixel SA restores the outline, and DenseCRF aligns the thresholded mask with image edges (yellow: annotation). Maps are averaged over SA calls 10–20 and all heads.

32 and 16 pixels (two in the encoder, three in the decoder) and one at 8 pixels. We use the eleven layers at 8, 16 and 64 pixels, $\mathcal{B}$, and discard the 32-pixel layers, which hurt in every combination (Sec. 5.1).

3.3. Self-attention-guided layer weighting

Not every CA layer is equally useful. Fine 64-pixel layers respond to texture and edges, coarse layers to whole objects (see the supplementary material). Averaging them uniformly gives the noisy fine layers five of eleven votes. We instead weight each layer by how well the spatial grouping of its block agrees with a reference grouping of the image, adapting the agreement score of GoCA [7] to every denoising call.

Reference grouping. At call t we take the CA queries $F_t \in \mathbb{R}^{n_r \times d_F}$ of one decoder block at 32 pixels (heads concatenated, n_r=1024). Their pairwise similarity

$$P_t = \mathrm{softmax}\big(F_t F_t^\top / \sqrt{d_F}\big) \in \mathbb{R}^{n_r \times n_r} \tag{3}$$

is a pseudo-affinity: entry (i, j) is large when positions i and j carry similar decoder features.

Agreement score. For each CA layer $b \in \mathcal{B}$, let $S_{b,t} = \frac{1}{H}\sum_h A^{\mathrm{sa}}_{b,t,h}$ be the head-averaged SA matrix of the *same* transformer block. We resize it to $n_r \times n_r$ as GoCA does (bilinear along keys, nearest along queries) and write the result, with rows indexed by keys, as $\tilde{S}_{b,t}$.[1] The layer's weight is its normalized, non-negative agreement with the reference:

$$w_{b,t} = \frac{\big[\langle \tilde{S}_{b,t}, P_t\rangle\big]_+}{\sum_{b'\in\mathcal{B}}\big[\langle \tilde{S}_{b',t}, P_t\rangle\big]_+}, \qquad \langle S, P\rangle = \sum_{i,j} S_{ij}P_{ij}, \tag{4}$$

where $[\cdot]_+ = \max(\cdot, 0)$. A block whose self-attention groups the image like the reference receives a large weight, and the weights at each call sum to one.

Semantic seed. The seed sums the weighted class-word maps over layers and calls at 64×64:

$$c = \mathrm{norm}\left(\frac{1}{31}\sum_{t=0}^{30}\sum_{b\in\mathcal{B}} w_{b,t}\, U_{64}(a_{b,t})\right) \in [0,1]^{64\times64}, \tag{5}$$

where U_{64} resizes bilinearly to 64×64 and norm rescales to $[0, 1]$ (min–max). On average, the weights place 84% of the mass on the 8- and 16-pixel layers (Sec. 5.1).

3.4. Coarse-to-fine self-attention propagation

The seed c marks the most class-specific parts of the object and leaves the rest weak (Fig. 2). Propagating it with SA completes the object. Instead of propagating with fine 64-pixel SA alone, we pass the map through SA at 16, 32 and 64 pixels in turn, using the middle SA calls $t \in \mathcal{T} = \{10, \ldots, 20\}$, which were selected for the fine-only baseline (supplementary material). Maps are flattened when multiplied by an SA matrix. For each call:

$$r_t = \bar{A}^{16}_t\, D(c) \qquad \in \mathbb{R}^{16\times16} \quad \text{(coarse envelope)} \tag{6}$$

$$q_t = \bar{A}^{32}_t\, U(r_t) \qquad \in \mathbb{R}^{32\times32} \quad \text{(finer proposal)} \tag{7}$$

$$z_t = \mathrm{cap}(q_t;\, r_t) \qquad \in \mathbb{R}^{32\times32} \quad \text{(capped proposal)} \tag{8}$$

and the final map averages the fine propagation of the capped maps over calls:

$$y = \frac{1}{|\mathcal{T}|}\sum_{t\in\mathcal{T}} \bar{A}^{64}_t\, U(z_t) \in \mathbb{R}^{64\times64}. \tag{9}$$

$D(c)$ averages each 4×4 block of the seed. Coarse SA in Eq. (6) spreads it over whole regions that the network groups together. Eq. (7) redistributes this evidence at 32 pixels, where boundaries are sharper but affinities can also reach look-alike background.

[1] GoCA indexes rows by queries. On our data the two orientations give seeds with Pearson correlation 0.99999 and the same results (supplementary material).

One-sided cap. Every coarse cell g has four children $\mathcal{C}(g)$ at the next resolution. Because bilinear upsampling preserves local means, the parent's value r_g is the mean evidence that the coarse level assigns to those children. The cap lowers any group whose mean exceeds that budget and leaves all other groups unchanged:

$$\mathrm{cap}(q;r)_k = q_k \cdot \min\Big(1,\ \frac{r_{g(k)}}{\bar{q}_{g(k)}}\Big), \qquad \bar{q}_g = \frac{1}{4}\sum_{k\in\mathcal{C}(g)} q_k, \tag{10}$$

where $g(k)$ is the parent of child k (a small $\epsilon{=}10^{-12}$ is added to $\bar{q}_g$). Within a capped group all children are scaled by the same factor, so their relative values are kept. The removed mass is discarded, not moved elsewhere. Equation (10) is the closest map to q, in generalized KL divergence, that satisfies the upper bounds $\sum_{k\in\mathcal{C}(g)} z_k \le 4r_g$ (proof in the supplementary material). It treats the coarse level as an *upper* bound on where the object can be. The natural alternative, *exact conservation* ($z_k = q_k\, r_{g(k)}/\bar{q}_{g(k)}$ for every group), also forces children to fill a parent's budget, even when the coarse map wrongly made that parent confident. Sec. 5.3 shows that only the one-sided version helps. We cap the $16 \to 32$ step only; the 64-pixel step stays free so that fine SA can sharpen boundaries.

3.5. Mask prediction and boundary refinement

The map y is rescaled to $[0,1]$, upsampled bilinearly to image size and thresholded at a single global threshold τ, which is chosen on the training folds (Sec. 4; $\tau{=}0.55$). This gives a binary mask b. Because SA operates on a 64×64 grid, the boundary of b is blocky. DenseCRF [3] aligns it with the image by minimizing, over pixel labels $x_i \in \{0,1\}$,

$$E(x) = \textstyle\sum_i \psi_u(x_i) + \sum_{i<j}[x_i \neq x_j]\, k(i,j), \tag{11}$$
$$k(i,j) = w_1 \exp\Big(-\tfrac{\|p_i-p_j\|^2}{2\theta_\alpha^2} - \tfrac{\|I_i-I_j\|^2}{2\theta_\beta^2}\Big) + w_2 \exp\Big(-\tfrac{\|p_i-p_j\|^2}{2\theta_\gamma^2}\Big).$$

The unary term ψ_u comes from b. The pairwise term penalizes giving different labels to pixels i, j that are close in position p and similar in colour I (first kernel), or simply close (second kernel), so the mask boundary moves onto image edges. We use OVAM's fixed parameters ($w_1{=}10$, $\theta_\alpha{=}80$, $\theta_\beta{=}13$, $w_2{=}3$, $\theta_\gamma{=}3$, five iterations) [6]. No ground truth enters any step.

4. Experimental setup

Benchmarks and metric. We use the two benchmarks released with OVAM [6]: VOC-Sim, with 490 annotated images of the 20 VOC classes from prompts "a photograph of a [class]", and COCO-Cap, with 707 annotated images generated from COCO captions, often cluttered and sometimes without the class name in the caption. We regenerate each image with Stable Diffusion 1.5 from its released prompt and seed (30 steps, guidance 7.5) to record attention. The metric is the mean over 20 classes of the class IoU, pooled over all images of the class.

Fold-safe selection. VOC-Sim is the only selection set. Its images are split once into five class-stratified folds. A design choice and its threshold are chosen on four folds and scored on the fifth. Pooling the five held-out folds gives the out-of-fold (OOF) estimate, and *forced OOF* scores a named variant in every fold. A change was promoted when it gained at least 0.010 training-fold mIoU (or 0.005 with 12 of 20 classes improved and no class losing more than 0.05) and won at least four of five folds. COCO-Cap is scored once, with the method and threshold frozen on VOC-Sim. Every comparison in Secs. 5.1–5.5 is between cells of one run on the same generated images. Paired intervals are 95% bootstrap intervals over images (20,000 resamples).

Selection status. The CA scales, the SA module, the two-pass baseline, DenseCRF and the SA call window were promoted by this rule. Two components were adopted after the rule had retained a simpler reference. The self-attention-guided layer weights won 4/5 folds and transferred, but the multi-candidate screen kept uniform weights. The capped hierarchy won 5/5 folds against both references, but one class (sheep, Sec. 5.5) lost more than the 0.05 guard. The supplementary material reports the registered outcomes in full.

5. What each stage contributes

Figure 2 shows the pipeline on images chosen by a fixed rule, and Table 1 measures each decision against the previous one in the same run. We now explain each step.

5.1. Seed: which CA scales, and how to weight them

CA is the only source of class identity, so the seed sets an upper bound on everything downstream. Table 2 (top) shows that no single CA scale suffices. The 16-pixel modules are the strongest alone, 8-pixel modules localize coarsely, and 64-pixel CA alone is nearly unusable (0.17 mIoU) because it fires on texture (see the supplementary seed gallery). The 32-pixel modules hurt in every combination: removing them raises OOF mIoU from 0.665 to 0.691 in 4/5 folds (paired $+0.022$, CI $[+0.015, +0.029]$). Restricting CA heads or calls did not survive once crossed with this choice.

Uniform averaging still lets the noisy 64-pixel modules contribute five of eleven votes. The self-attention-guided weights of Eq. (4) are not learned, yet they shift 84% of the weight to the 8- and 16-pixel modules, most strongly



Figure 2. **From seed to mask.** Columns are chosen by a fixed rule, without inspecting any image: the VOC-Sim classes at the best, upper-third, lower-third and worst class IoU of the final pipeline, and the COCO-Cap classes at the best, median and worst, each shown by its median-IoU image. Rows: generated image; CA seed c (Eq. 5); after coarse 16-pixel SA (r); after the cap and fine 64-pixel SA (y); final mask after DenseCRF (blue) with the annotation outline (yellow). The VOC dining table shows the main failure mode: SA groups the table with its chairs.

at early, noisy calls (supplementary material). Interestingly, the weighting *lowers* the CA-only score (0.714→0.705) but raises the final mask from 0.817 to 0.823 (4/5 folds) and COCO-Cap from 0.644 to 0.662 (Table 2, bottom). A seed is good if SA can propagate it, not if it is the best mask on its own. This is the first sign that coarse structure is the right base.

5.2. Envelope: why start from coarse self-attention

SA at different scales groups pixels differently. Figure 3 (left, middle) compares propagating the same seed through 16-pixel SA only, 64-pixel SA only, and from 16 to 64 pixels. Coarse SA keeps recall high as the threshold rises (0.73 at τ=0.70 on VOC-Sim, against 0.61 for fine SA), because it assigns whole regions, not textures, to the object. Its precision is low, however: blocks along the boundary are all-or-nothing, and alone it scores only 0.723 OOF. Fine SA is precise but loses object parts at high thresholds. Going from coarse to fine keeps most of the coarse recall at nearly fine precision. On cluttered COCO-Cap, where fine SA leaks into look-alike textures, it already beats fine-only propagation (0.687 vs. 0.660 frozen). On VOC-Sim, however, it falls short (0.802 vs. 0.810 OOF). The coarse envelope is useful, but on its own it lets the finer levels claim too much. The cap addresses this.

5.3. Detail within the envelope: cap, not conservation

How strongly should the coarse level constrain the next? Table 3 answers this in one run. The plain hierarchy (no constraint) loses to fine-only propagation on VOC-Sim. Forcing exact mass conservation at both transitions is far worse (0.759 raw, −5 points). Conservation at 16 → 32 alone, or conservation followed by fine completion, stays below fine-only after DenseCRF. The one-sided cap is the only variant that beats both fine-only propagation and the plain hierarchy in every fold: 0.830 after DenseCRF, +0.012 over the plain hierarchy (CI $[+0.006, +0.013]$), with 0.697 on COCO-Cap. A second cap at 32 → 64 adds nothing, which confirms that fine SA should refine boundaries freely. Re-adding completion on top of the cap re-introduces false positives and erases the COCO-Cap gain.

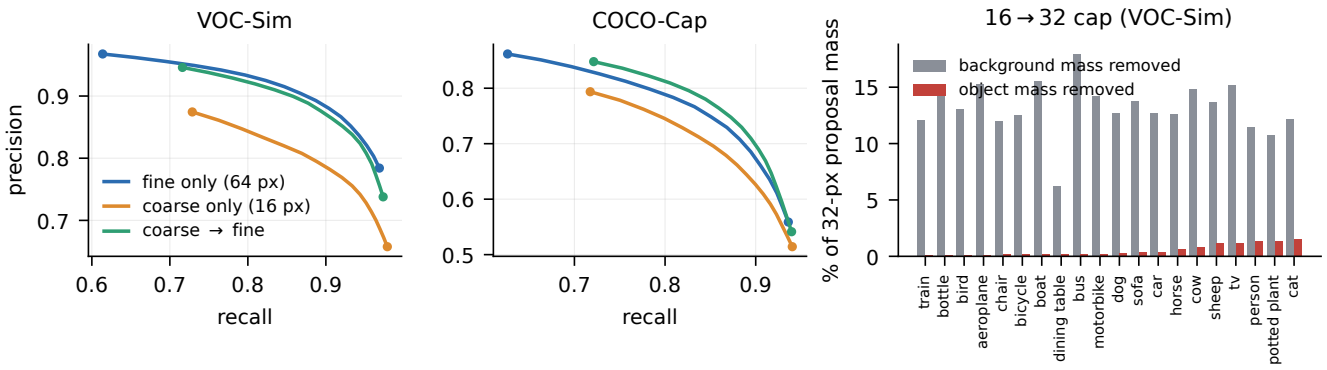


Figure 3. **Coarse SA gives the envelope; the cap trims only background.** *Left, middle:* pooled precision and recall over the threshold grid (0.30 at the high-recall end to 0.70) for one seed propagated by 64-pixel SA only, 16-pixel SA only, or 16→64 pixels (two passes per level, same run). Coarse SA keeps recall at high thresholds and coarse-to-fine combines it with fine precision. *Right:* share of the 32-pixel proposal's mass that the cap removes inside the annotated object (red) and in the background (grey), per VOC-Sim class (ground truth is used only for this diagnostic). Overall 13% of background mass and 0.5% of object mass are removed.

no 32-px CA	SA 64 px	2 passes	DenseCRF	calls 10–20	SA-guided weights	coarse-to-fine + cap	VOC	Δ	COCO
							.665		–
✓							.691	+.026	–
✓	✓						.789	+.033[a]	–
✓	✓	✓					.800	+.012	.625
✓	✓	✓	✓				.813	+.014	.637
✓	✓	✓	✓	✓			.817	+.004	.644
✓	✓	✓	✓	✓	✓		.823	+.006	.662
✓	✓	✓	✓	✓	✓	✓	**.829**	+.006	**.696**

Table 1. **The pipeline, one decision at a time.** VOC: OOF mIoU; Δ: gain over the previous row *measured in the same run* (same generated images); COCO-Cap: VOC-frozen transfer. The first three rows come from early one-axis screens at a fixed threshold 0.45 (CA only, then one SA pass); [a]against one pass through all sixteen SA modules (.756). Later rows select thresholds inside the training folds. Runs differ by at most 0.0003 on a shared reference.

CA resolutions (px) 8	16	32	64	CA seed only, VOC-Sim mIoU@.45	AP
✓				.563	.846
	✓			.649	.906
		✓		.300	.822
			✓	.173	.560
✓	✓	✓	✓	.666	.922
✓	✓			.693	.917
✓	✓		✓	**.696**	.913

Layer weights $w_{b,t}$	CA only	full pipeline	COCO
uniform	.714	.817	.644
SA-guided (Eq. 4)	.705	**.823**	**.662**

Table 2. **Building the semantic seed.** *Top:* CA-only mIoU at threshold 0.45 and average precision for resolution subsets (all calls and heads; full VOC-Sim, descriptive). Each scale alone is weak and 64-px CA is nearly unusable alone, but 8, 16 and 64 px are complementary; dropping the 32-px modules raises out-of-fold (OOF) mIoU from 0.665 to 0.691 (4/5 folds). *Bottom:* module weighting, forced OOF in one run, followed by fine-only propagation and DenseCRF. SA-guided weighting lowers the CA-only score yet improves the final mask in 4/5 folds and on COCO-Cap: it favours modules whose spatial structure SA can propagate.

Why does the one-sided form work? The coarse parent is a reliable *upper bound* on where the object can be, but not a reliable lower bound. The cap binds in 58% of 2×2 groups yet removes only 7.6% of the proposal mass. As Fig. 3 (right) shows, that mass is almost all background: 13% of the background mass but 0.5% of the object mass, and at most 1.6% for any class. Conservation instead makes children inherit the budget of every parent, including parents that the blurry coarse map wrongly made confident, and it must spend that budget somewhere. Figure 4 shows the effect. Conservation distorts the mask wherever a parent's budget is wrong (COCO cat: IoU 0.58, against 0.81 with the cap), whereas the cap removes mass only where the coarse level disagrees.

5.4. Boundary: what DenseCRF adds

SA works on a 64-pixel grid, so even a good map has blocky, slightly dilated borders (Fig. 5). DenseCRF uses the image itself: the pairwise kernel of Eq. (11) prefers equal labels for nearby pixels of similar colour, so mask boundaries snap to edges. With fixed parameters it improves 278 VOC-Sim images and worsens 97, and raises mIoU by 0.013 (Table 4). Its effect depends on what it is given. On fine-only maps it mainly adds recall and barely helps COCO-Cap

Variant	coarse→fine	conserve (=) 16→32	conserve (=) 32→64	cap (≤) 16→32	cap (≤) 32→64	compl.	wins	mIoU VOC raw	mIoU VOC + CRF	mIoU COCO + CRF
References										
fine only (64 px)							–	.8098	.8231	.6622
plain hierarchy	✓						1/5	.8061	.8184	.6925
Two-sided: children must carry exactly the parent's mass										
conservation	✓	✓					2/5	.8034	.8185	.7038
conservation	✓		✓				0/5	.7894	.8135	.7022
conservation	✓	✓	✓				0/5	.7592	.7917	.6851
conservation + completion	✓	✓	✓			✓	2/5	.8037	.8204	**.7063**
One-sided: children may not exceed the parent's mass										
cap (ours)	✓			✓			**5/5**	**.8157**	**.8304**	.6966
cap	✓			✓	✓		4/5	.8151	.8290	.6964
cap + completion	✓			✓		✓	3/5	.8131	.8209	.6647

Table 3. **How the coarse level should constrain the finer one.** All rows share one generation of the 490 VOC-Sim and 707 COCO-Cap images, the same seed and SA calls 10–20. *Coarse→fine*: propagation through SA at $16 \to 32 \to 64$ px (otherwise 64-px SA only); CRF: DenseCRF. *Conserve* forces each 2×2 child group to carry exactly its parent's mass (closed-form KL projection). *Cap* only removes mass above the parent (Eq. 10). *Compl.*: fine-SA completion re-adds positive residuals. VOC: forced OOF mIoU; *wins*: held-out folds in which a row beats fine-only after DenseCRF; COCO-Cap: VOC-frozen threshold. Forcing parent mass onto children loses up to 5 VOC points. Capping at $16 \to 32$ is the only variant that beats both references in every fold (vs. plain hierarchy: $+0.012$ after DenseCRF, paired 95% CI $[+0.006, +0.013]$).

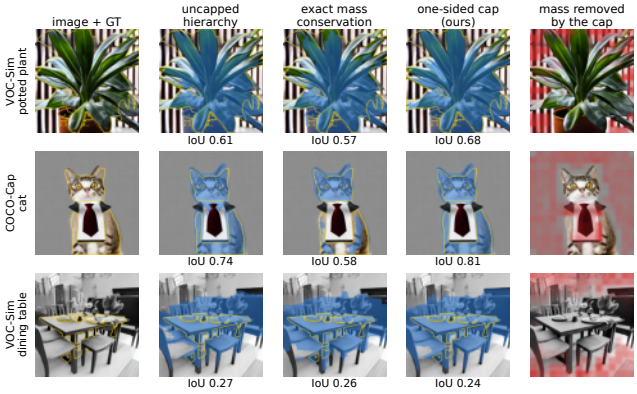


Figure 4. **Cap versus conservation.** Images at the 95th percentile of the cap's IoU gain over the plain hierarchy on VOC-Sim and COCO-Cap, and at the 5th percentile on VOC-Sim (fixed rule). Masks at each variant's VOC-selected threshold. Right: mass removed by the cap (red). The cap removes background mass that the coarse level does not support; conservation spreads mass into it. When the coarse level groups the table with its chairs (bottom), no variant can separate them.

(+0.002, with 243 images worse). On capped coarse-to-fine maps it raises precision, helps COCO-Cap six times more (+0.014, 18 of 20 classes), and turns 427 images better against 105 worse. Once the cap has removed far-away mass, the remaining errors lie near boundaries, where appearance is informative. DenseCRF fails when appearance misleads: a chair's thin legs against a similar background (Fig. 5, bottom).

coarse-to-fine	CRF	VOC-Sim mIoU	VOC-Sim P / R	VOC-Sim img ↑/↓	COCO-Cap mIoU	COCO-Cap P / R	COCO-Cap img ↑/↓
		.811	.880 / .904		.660	.708 / .882	
	✓	.824	.883 / .916	291/132	.662	.703 / .889	354/243
✓		.817	.875 / .915		.682	.731 / .889	
✓	✓	**.830**	**.890** / .910	278/97	**.696**	**.747** / .886	427/105

Table 4. **What DenseCRF adds, and why it pairs with the cap.** Full-VOC mIoU, pooled precision/recall and the number of images whose IoU rises/falls by more than 0.005 when DenseCRF is added (same run; thresholds from VOC-Sim, frozen for COCO-Cap). On fine-only maps DenseCRF mainly recovers recall and barely helps COCO-Cap (+0.002). On capped coarse-to-fine maps it raises precision and helps COCO-Cap six times more (+0.014, 18 of 20 classes): with far-away mass removed, the remaining errors lie near object boundaries, where appearance decides.

5.5. What the hierarchy changes: calibration across images

The cap barely changes the *ranking* of pixels within an image: raw average precision on COCO-Cap is unchanged (-0.0006). If each image could choose its own threshold (an oracle that uses ground truth), fine-only and coarse-to-fine maps would tie (0.896 vs. 0.895 after DenseCRF, Fig. 6). The gain lies in *calibration*: after the cap, one global threshold fits many more images. Removing background mass makes small objects stand out, so the best global threshold moves from 0.425 to 0.55, and at that threshold the curve peaks higher on both benchmarks.

This also explains the one class that loses. Sheep images



Figure 5. **DenseCRF snaps the mask to image edges.** Images at the 90th percentile of the DenseCRF gain on VOC-Sim and COCO-Cap, and at the 10th percentile on VOC-Sim (fixed rule).

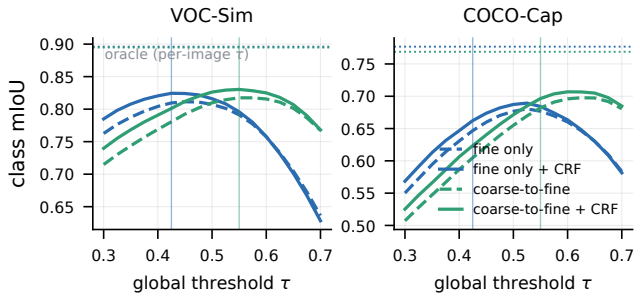


Figure 6. **The hierarchy improves calibration, not ranking.** Class mIoU against the global threshold for fine-only and coarse-to-fine maps, raw (dashed) and with DenseCRF (solid), on all images of one run. Vertical lines: VOC-selected thresholds (0.425 and 0.55). Dotted: per-image best threshold (oracle, uses ground truth), which ties the two maps.

are often close-ups that fill most of the frame and prefer a low threshold (0.35), at which the capped map leaks into the surroundings. At the shared threshold of 0.55 sheep IoU falls by 0.09, although at a matched threshold the capped map is better on sheep and the cap removes only 1.1% of sheep pixels. None of six label-free per-image thresholding rules improved the capped pipeline (supplementary material). The loss is a calibration trade-off of a single global threshold, not a failure of the propagation.

6. Comparison with published methods

Table 5 places the pipeline beside the class-word results reported by OVAM on the same benchmarks. With class words only and no training, it exceeds OVAM by 0.125 on VOC-Sim and 0.114 on COCO-Cap. It also exceeds DatasetDM, which trains an extra decoder, on both benchmarks. Most of the COCO-Cap margin over our fine-only pipeline comes from the capped hierarchy: cluttered scenes benefit most from a coarse envelope. Released baseline masks scored under our evaluation are in the supplementary material.

Method	VOC-Sim	COCO-Cap
Training-free, class word (published in OVAM, Table 1)		
DAAM [11]	.662	.484
Attn2Mask	.687	.550
OVAM [6]	.704	.582
With extra training (published in OVAM, Table 1)		
Grounded Diffusion	.621	.502
DatasetDM	.803	.593
Ours, training-free, class word		
fine-only propagation + DenseCRF	.823	.662
coarse-to-fine + cap + DenseCRF	**.829**	**.696**

Table 5. **Comparison with published class-word results** on the benchmarks released by OVAM [6]; optimized-token rows are excluded. Ours: forced out-of-fold VOC-Sim, frozen COCO-Cap. We regenerate each image from its released prompt and seed, so cross-block differences are indicative rather than paired.

7. Discussion and limitations

Negative results. We also registered 45 comparisons of alternative operators, such as random walks, graph sharpening and re-seeding between scales (supplementary material). Several improved the raw map, but apart from the SA call window none survived DenseCRF. The capped hierarchy is the largest change that survived refinement in every fold and transferred.

Failure modes and scope. The pipeline inherits SA's notion of an object. When co-occurring objects share coarse affinity (the dining table and its chairs, Figs. 2 and 4) or the seed is weak (COCO-Cap person, our weakest class at 0.28 IoU), propagation fills the wrong region, and frame-filling objects lose under one global threshold (Sec. 5.5). Both benchmarks contain generated images, regenerated from released prompts and seeds, and all choices are made on VOC-Sim with one backbone. Two components were adopted after the registered rule kept a simpler reference (Sec. 4); their gains hold in 4/5 and 5/5 held-out folds and transfer, but they are not pre-registered promotions.

8. Conclusion

Training-free segmentation from diffusion attention works best when each scale does what it is good at: coarse attention decides where the object is, finer SA refines its boundary within the mass the coarse level allows, and DenseCRF aligns it with the image. The one-sided cap is what makes this division of labour work.

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